INTEGRATED APPLICATION ? PROGRESS REPORT ? NOVEMBER 2026

# Flood Risk & Disaster Management

*A connected platform for flood monitoring, data review, scenario modelling and response follow-up.*

| PREPARED BY | REPORTING PERIOD | STATUS |
|---|---|---|
| Flood Risk & Disaster Management | November 2026 | Active development |


REPORT SUMMARY

The application brings operational monitoring and flood research into one workflow. Users can review weather and river context, explore district-level risk, organize reports and incidents, inspect prepared datasets, and run or compare georeferenced flood-model results. The interface keeps data provenance visible and distinguishes live or rule-based screening from model outputs that still need scientific validation.


## Main application features

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| 01 | **Live monitoring & decision support**<br>A dashboard brings current weather and GloFAS modelled river discharge together with map-based monitoring. National GIS screening covers 126 Pakistan districts and supports district, coordinate and catchment exploration. Citizen reports, incidents, recent activity and alert context help users review signals and follow up. |
| 02 | **Data, provenance & exposure**<br>The research workspace organizes Marala rainfall and river-flow records with source references, quality checks and visual summaries. A Trimmu/Jhang exposure view combines population, roads, buildings, facilities, settlements and historical flood references to support local risk review. |
| 03 | **Scenario simulation & result review**<br>Users can define or load scenarios, register scientific inputs, prepare a model domain and bind a matching external engine project. HEC-RAS 2D and LISFLOOD-FP workflows support computation and publication of timestamped geospatial outputs. Map playback, hydrographs and side-by-side result comparison help inspect model behaviour. |

## Progress at a glance

| DELIVERED | DEMONSTRATED | IN PROGRESS |
|---|---|---|
| Monitoring dashboard, map exploration, report and incident workflows, data portal, scenario and result review. | Public-data screening and HEC-RAS cases have been computed and published for interactive review; source checks support reproducibility. | Field observations, calibration and independent validation are the main work ahead; results remain exploratory for flood prediction. |

READINESS, EVIDENCE & NEXT STEPS

# Progress & next priorities

The platform now connects monitoring, research inputs and model-result review in a single user-facing experience. The work has moved from interface foundations to runnable, traceable workflows; the remaining focus is on strengthening the evidence behind operational and hydraulic conclusions.

## Current readiness

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| **Monitoring** | Live weather and modelled discharge are presented with an explicit distinction between modelled flow and measured water level. Rule-based screening is the primary operational risk method; the bundled ML artifact is not enabled for production use. |
| **Model outputs** | Public-data two-dimensional cases and upstream routing experiments demonstrate the preparation, compute, publish and map-review workflow. Scenario and input checks help keep runs tied to the data and engine configuration used. |
| **Validation** | Hydraulic cases are not yet calibrated or independently validated. Channel-bed depths are estimated, observed records are sparse, and a withheld routing skill target has not been met. Published maps are exploratory results, not confirmed flood forecasts. |

## Next development priorities

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| **1 Improve observations** | Obtain reviewed gauge time series, measured channel geometry and event-specific flood extents for the selected study areas. |
| **2 Calibrate & validate** | Re-run hydrology and hydraulic cases against independent observations; document skill scores, numerical checks and uncertainty before widening claims. |
| **3 Complete data handoffs** | Keep source coverage, dates, licensing and quality notes visible, and connect approved authority-supplied emergency resources when available. |
| **4 Harden operations** | Continue improving provider-failure handling, refresh visibility, reproducible scenario setup and clear separation of operational screening from research tools. |

## Overall contribution

This work provides a shared path from checking flood conditions to reviewing data, exploring model scenarios and organizing response follow-up. The main features are in place for integrated use; expanding dependable observations and completing independent validation are the key steps toward stronger forecasting confidence.